Inverse Trigonometric Functions

Exam code: 9MA0

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Inverse trigonometric functions

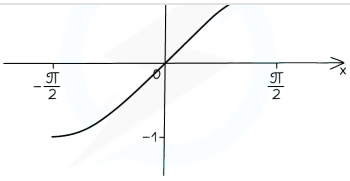
What are the inverse trigonometric functions?

- These functions are the **inverse functions** of **sin**, **cos** and **tan**
  - **sin (arcsin x) = x**
  - **cos (arccos x) = x**
  - **tan (arctan x) = x**
- The domains of **sin**, **cos**, and **tan** must first be restricted to make them one-to-one functions (only one-to-one functions have inverses)

What are the restricted domains?

- domain of **sin x** is restricted to $-\pi/2 \le x \le \pi/2$ ($-90° \le x \le 90°$)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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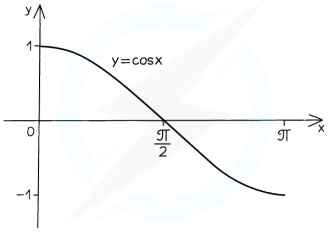
- domain of **cos *x*** is restricted to **0 ≤ *x* ≤ π** (**0° ≤ *x* ≤ 180°**)



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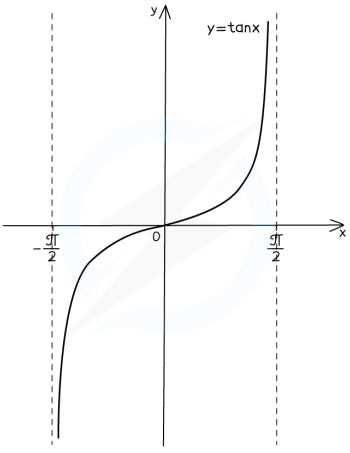
- domain of **tan *x*** is restricted to **−π/2 < *x* < π/2** (**−90° < *x* < 90°**)



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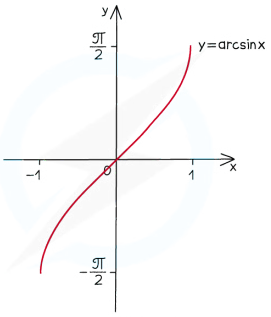
How do I sketch the graph of y = arcsin x?

- The graph of ***y* = arcsin *x*** looks like this:



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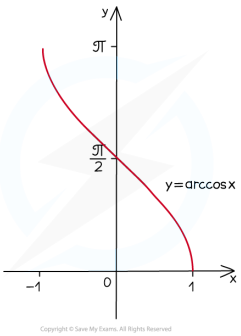
How do I sketch the graph of y = arccos x?

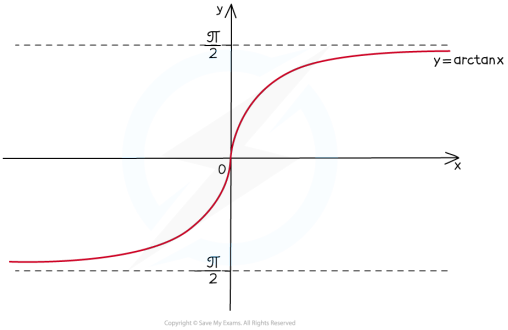
- The graph of $y = \arccos x$ looks like this:



- *the* **domain** is $-1 \le x \le 1$
- the **range** is $0 \le \arccos x \le \pi$ ($0° \le \arccos x \le 180°$)

How do I sketch the graph of y = arctan x?

- The graph of $y = \arctan x$ looks like this:



- the **domain** is $x \in \mathbb{R}$ (ie arctan x is defined for *all* real number values of x)
- the **range** is $-\pi/2 < \arctan x < \pi/2$ ($-90° < \arctan x < 90°$)
- horizontal **asymptotes** at $y = -\pi/2$ and $y = \pi/2$

Examiner Tips and Tricks

- Make sure you know the shapes of the graphs for **sin**, **cos** and **tan.**
- As inverses, the graphs of **arcsin**, **arccos** and **arctan** are reflections of **sin**, **cos** and **tan** in the line $y = x$.

Worked Example

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